

Mithra Kancheti

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EDUCATION

Purdue University

Aug. 2022 – May. 2025

Double BS in Computer Science and Data Science, Minor in Mathematics

Core Courses: Machine Learning, Numerical Methods, Systems Programming, Statistical Theory

Certification: SAS 9.4 Base Programming Specialist

Teaching: MA 265: Linear Algebra TA

EXPERIENCE

Kaiser Permanente - Software Engineer

Feb. 2026 – Present

- Building, and maintaining data pipelines across multiple projects between Databricks, Azure, Oracle Server, etc. These support the most critical projects, satisfying intense business and regulatory scrutiny.
- Leading internal reporting refinement initiatives. Directly corresponding with business leaders, nurses, and hospital program managers to set milestones and convey timelines.

Kaiser Permanente - Jr Software Engineer

Jun. 2025 – Feb. 2026

- Developer for internal tools to automate and optimize manual workflows, collaborating with technical and non-technical stakeholders to create and adhere to shifting specifications.
- Developing Github Actions and Jenkins integrations to ensure the longevity and reliability of my products.
- Rewrote core database schemas to ensure dimension normalization, increasing data integrity and query performance.

Purdue University - Linear Algebra Teaching Assistant

Dec. 2022 – May 2023

- Assisted students in office hours, providing feedback and homework help.
- Organized and led study sessions, drafting materials and plans to maximize productivity.

PERSONAL

GiftGen | AWS, Terraform, Kubernetes

- Created a website for users to generate objects using a custom Text-to-3D-model inference endpoint using Microsoft Trellis. Inference runs as a containerized workload on Modal.
- Using EKS, S3, RDS, Cloudwatch, and Cognito to run a resilient and scalable backend. Separate Dev and Prod environments are managed with Terraform, and CodePipeline with ArgoCD enables PR based promotion.
- Automatically handling DNS resolution using Terraform's Cloudflare integration, ensuring infrastructure updates are instantly visible.

Semantic Search Media Server | Python, LLM, React, SQL, Docker

- Built a selfhosted, LLM powered system that automatically generates semantic tags for personal media files (text, images, and videos) to enable natural language search.
- Implemented entropy-based filtering to identify and downscale low-complexity images, optimizing speed and resource usage during model inference.

Song Performance Prediction | Python, Pytorch, Postgres, Scikit-learn

- Designed various regression models and a neural network to perform prediction of a song's performance based on track and artist attributes collected from various streaming services.
- Achieved a final accuracy of 90% with an R^2 of 0.98
- Collaborated in a team of 8 and used agile techniques such as stand-ups and Kanban boards to manage progress.

Personal Server | Proxmox, Docker, Grafana, Networking

- Set up Proxmox VE to act as a hypervisor for various services including Jupyter Server, Ollama, PocketBase, etc. Acts as my development backend when starting on projects.
- Utilizing Docker Swarm to manage containerization and extensibility of service stacks, integrated with Prometheus and Grafana to monitor the health of services.
- VLAN tagging VM traffic ensures a DMZ between my public services and my internal network.

TECHNICAL SKILLS

Languages: C/C++, Java, Python, R, JavaScript, Flutter, MATLAB, SQL, x86-64 Assembly, Cypher

Frameworks/Libraries: NumPy, Pandas, PyTorch, Pyspark, Playwright

Tools: PowerBI, Kubernetes, Docker, Terraform, Ghidra